

ECONOMETRIC THEORY AND PRACTICE I

Quiz 1 - Probability Review

3 questions | 10 minutes | 1 point each | No negative marking



Date: _____

Work independently. No notes or calculators.

No phones, devices, or AI tools.

Use the blank reverse side for calculations.

Keep the QR code and corner marks clear.

form ref: B7CS

STUDENT ID - 3 DIGITS

Write each digit, then fill its bubble.

	digit 1	digit 2	digit 3
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9	☐	☐	☐

Fill all 3 ID digits clearly. An unassignable ID receives zero.

RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. A random variable X equals 2 with probability 0.75 and equals 10 with probability 0.25. What is $E(X)$?

- A. 2

C. 4

E. 2.5

G. 8
- B. 10

D. 6

F. 1.5

2. Suppose $E(Y) = 2$, $\text{Var}(Y) = 2$, and $X = 3Y + 4$. Which ordered pair gives $(E(X), \text{Var}(X))$?

- A. (10, 34)

C. (10, 6)

E. (6, 18)

G. (22, 18)
- B. (10, 2)

D. (10, 18)

F. (10, 22)

3. Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.25$, $E(W) = 18$, and $E(W | D = 0) = 14$. In population B, $\Pr(D = 1) = 0.75$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

- A. 14

C. 18

E. 22

G. 30
- B. 22.5

D. 17

F. 26

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form ref: UF5C

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9	☐	☐	☐

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RESPONSE PANEL

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Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. Suppose $E(Y) = 3$, $\text{Var}(Y) = 3$, and $X = 2Y + 5$. Which ordered pair gives $(E(X), \text{Var}(X))$?

- A. (11, 37)

C. (17, 12)

E. (11, 12)

G. (11, 6)
- B. (11, 3)

D. (11, 17)

F. (6, 12)

2. Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.75$, $E(W) = 24$, and $E(W | D = 0) = 12$. In population B, $\Pr(D = 1) = 0.25$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

- A. 24

C. 28

E. 15

G. 12
- B. 16

D. 20

F. 7

3. A random variable X equals 3 with probability 0.75 and equals 15 with probability 0.25. What is $E(X)$?

- A. 3

C. 3.75

E. 9

G. 15
- B. 2.25

D. 6

F. 12

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form ref: P76Y

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8	☐	☐	☐
9	☐	☐	☐

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RESPONSE PANEL Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.25$, $E(W) = 22$, and $E(W \mid D = 0) = 16$. In population B, $\Pr(D = 1) = 0.75$. The conditional means $E(W \mid D = 0)$ and $E(W \mid D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?
- A. 16

B. 30

C. 40

D. 22

E. 34

F. 28

G. 20.5
2. A random variable X equals 4 with probability 0.25 and equals 12 with probability 0.75. What is $E(X)$?
- A. 9

B. 8

C. 12

D. 4

E. 6

F. 1

G. 10
3. Suppose $E(Y) = 4$, $\text{Var}(Y) = 4$, and $X = 3Y + 2$. Which ordered pair gives $(E(X), \text{Var}(X))$?
- A. (14, 36)

B. (14, 12)

C. (12, 36)

D. (14, 40)

E. (14, 38)

F. (14, 4)

G. (38, 36)

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form ref: THZ2

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5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐

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RESPONSE PANEL

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Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. A random variable X equals 6 with probability 0.25 and equals 14 with probability 0.75. What is $E(X)$?

- | | |
|-------|---------|
| A. 10 | B. 10.5 |
| C. 14 | D. 1.5 |
| E. 6 | F. 12 |
| G. 8 | |

2. Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.75$, $E(W) = 34$, and $E(W | D = 0) = 10$. In population B, $\Pr(D = 1) = 0.25$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

- | | |
|-------|---------|
| A. 18 | B. 42 |
| C. 34 | D. 10.5 |
| E. 16 | F. 10 |
| G. 26 | |

3. Suppose $E(Y) = 5$, $\text{Var}(Y) = 5$, and $X = 2Y + 3$. Which ordered pair gives $(E(X), \text{Var}(X))$?

- | | |
|-------------|-------------|
| A. (10, 20) | B. (13, 5) |
| C. (13, 20) | D. (13, 10) |
| E. (13, 29) | F. (13, 23) |
| G. (23, 20) | |

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form ref: PEY5

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3	☐	☐	☐
4	☐	☐	☐
5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐

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RESPONSE PANEL

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Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. Suppose $E(Y) = 3$, $\text{Var}(Y) = 3$, and $X = 4Y + 6$. Which ordered pair gives $(E(X), \text{Var}(X))$?

- | | |
|-------------|-------------|
| A. (18, 12) | B. (12, 48) |
| C. (54, 48) | D. (18, 84) |
| E. (18, 54) | F. (18, 3) |
| G. (18, 48) | |

2. A random variable X equals 8 with probability 0.75 and equals 20 with probability 0.25. What is $E(X)$?

- | | |
|-------|-------|
| A. 20 | B. 11 |
| C. 5 | D. 17 |
| E. 14 | F. 6 |
| G. 8 | |

3. Let D be 0 or 1 and W be hourly earnings in euros. In population A, $\Pr(D = 1) = 0.25$, $E(W) = 26$, and $E(W | D = 0) = 18$. In population B, $\Pr(D = 1) = 0.75$. The conditional means $E(W | D = 0)$ and $E(W | D = 1)$ are the same in both populations. What is $E(W)$ in population B, in euros per hour?

- | | |
|-------|---------|
| A. 24 | B. 18 |
| C. 50 | D. 42 |
| E. 26 | F. 37.5 |
| G. 34 | |